



Surrogate-Gated Diffusion and Foundation-Model Embeddings for Bayesian Materials Design

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Motivation

Diffusion priors fine-tuned by reinforcement learning propose crystals faster than an oracle can score them. A surrogate placed before the oracle only needs to rank candidates to decide which few deserve an evaluation.

3

diffusion priors

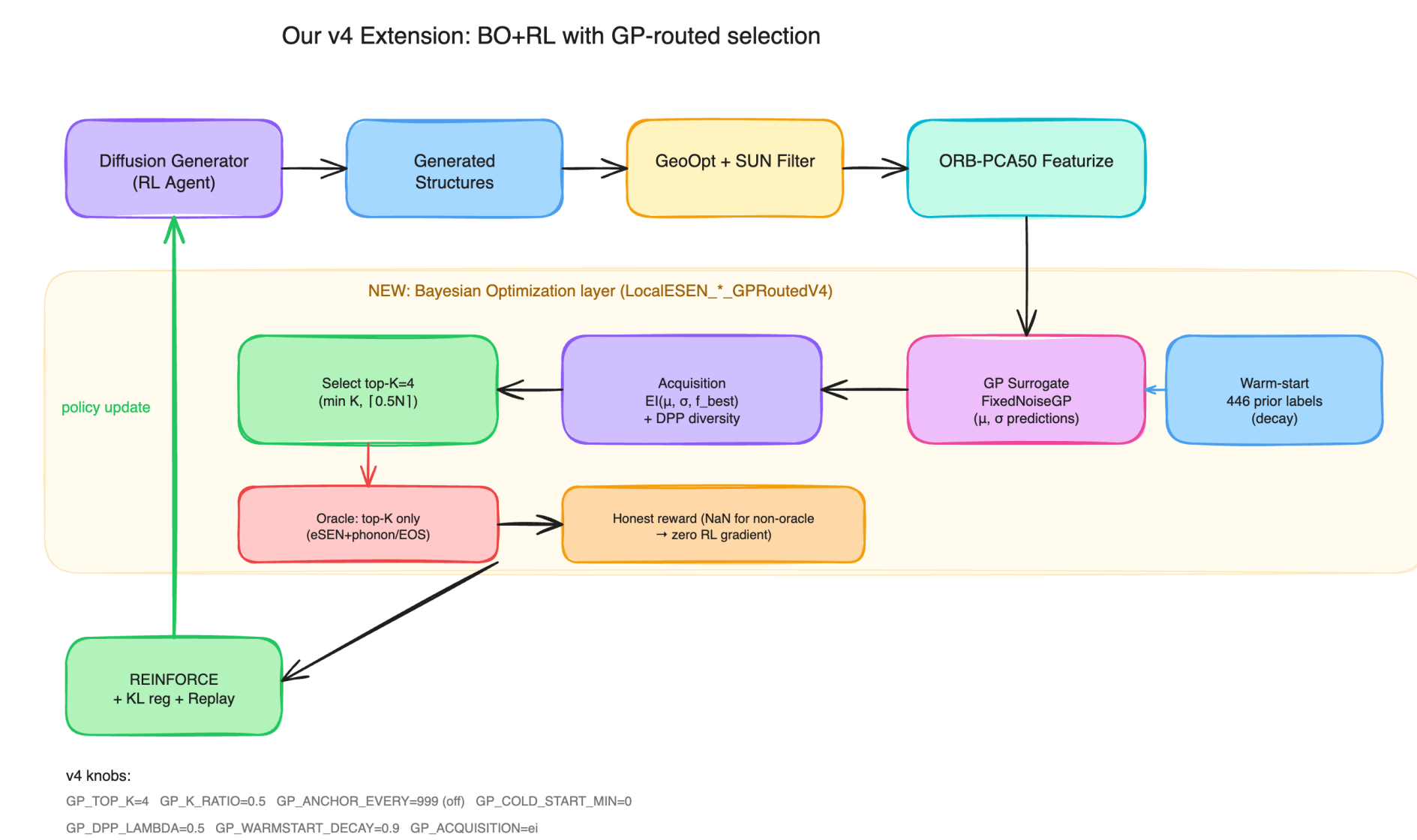
4

oracle calls per cycle

8,640

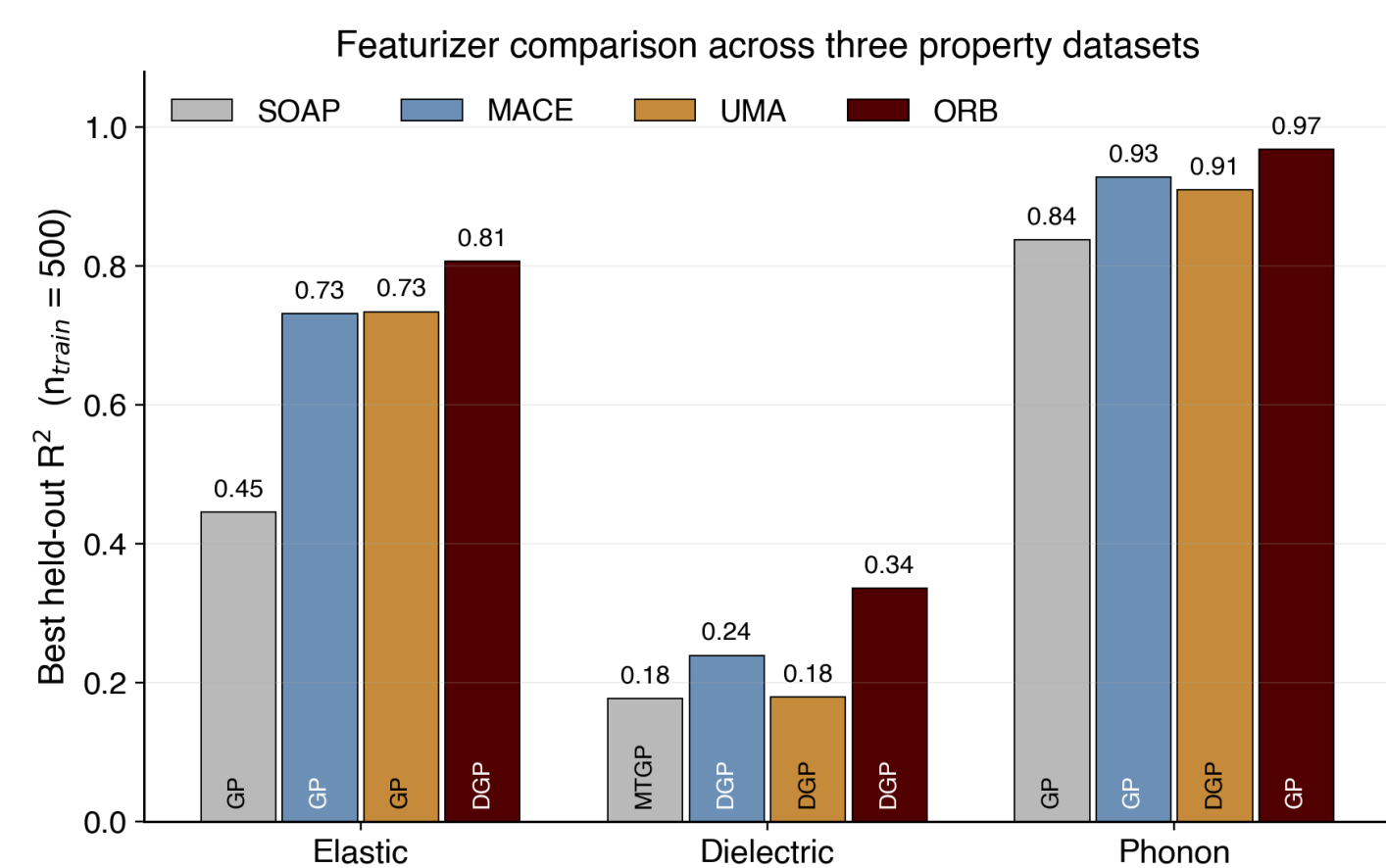
benchmark fits

The Gated Loop



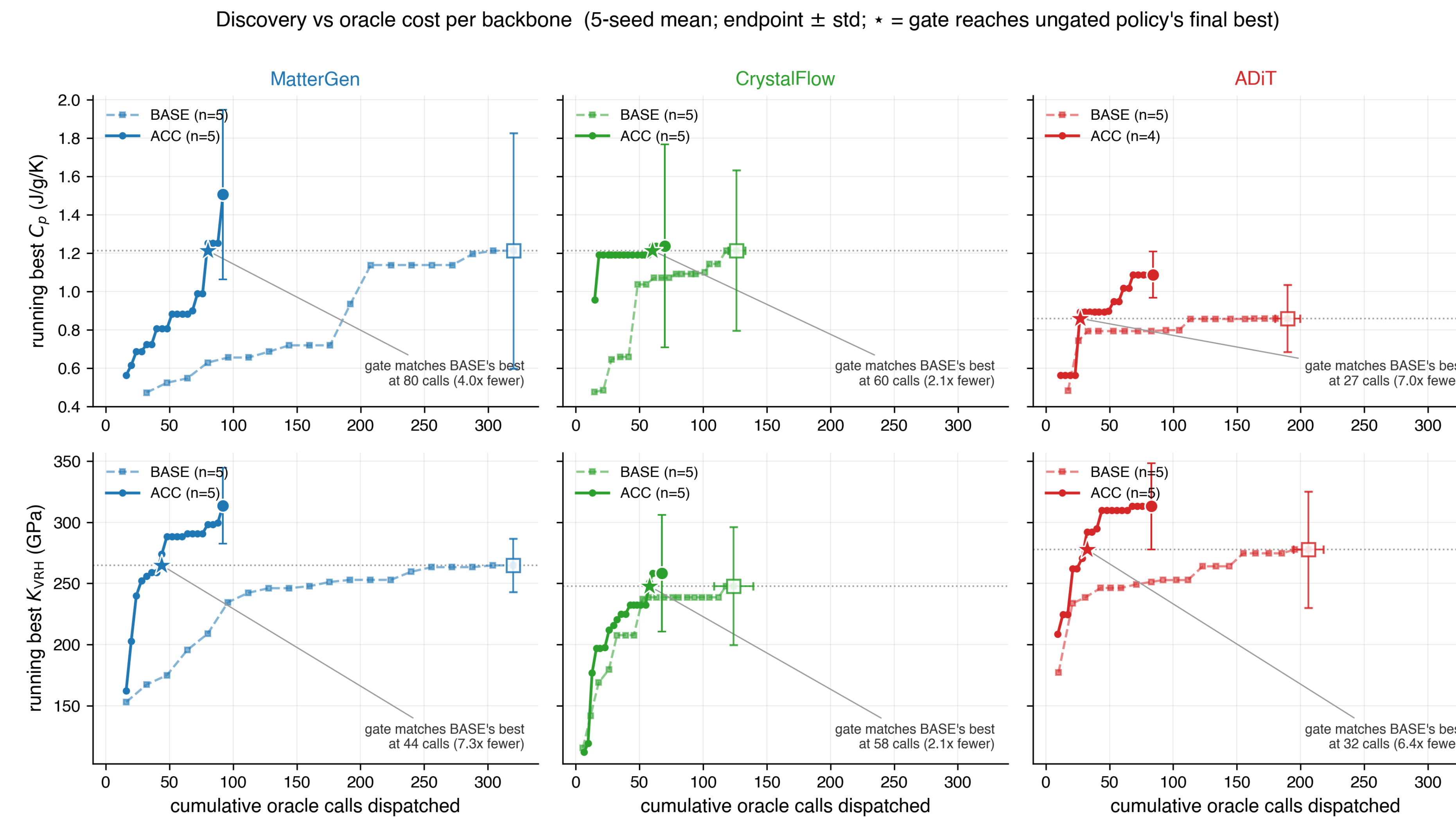
Stable, unique and novel survivors are featured with ORB and scored by a Gaussian process; Expected Improvement with a diversity term sends the top four to the oracle. The rest carry no reward, so the surrogate's choice is the only training signal.

Choosing the Surrogate



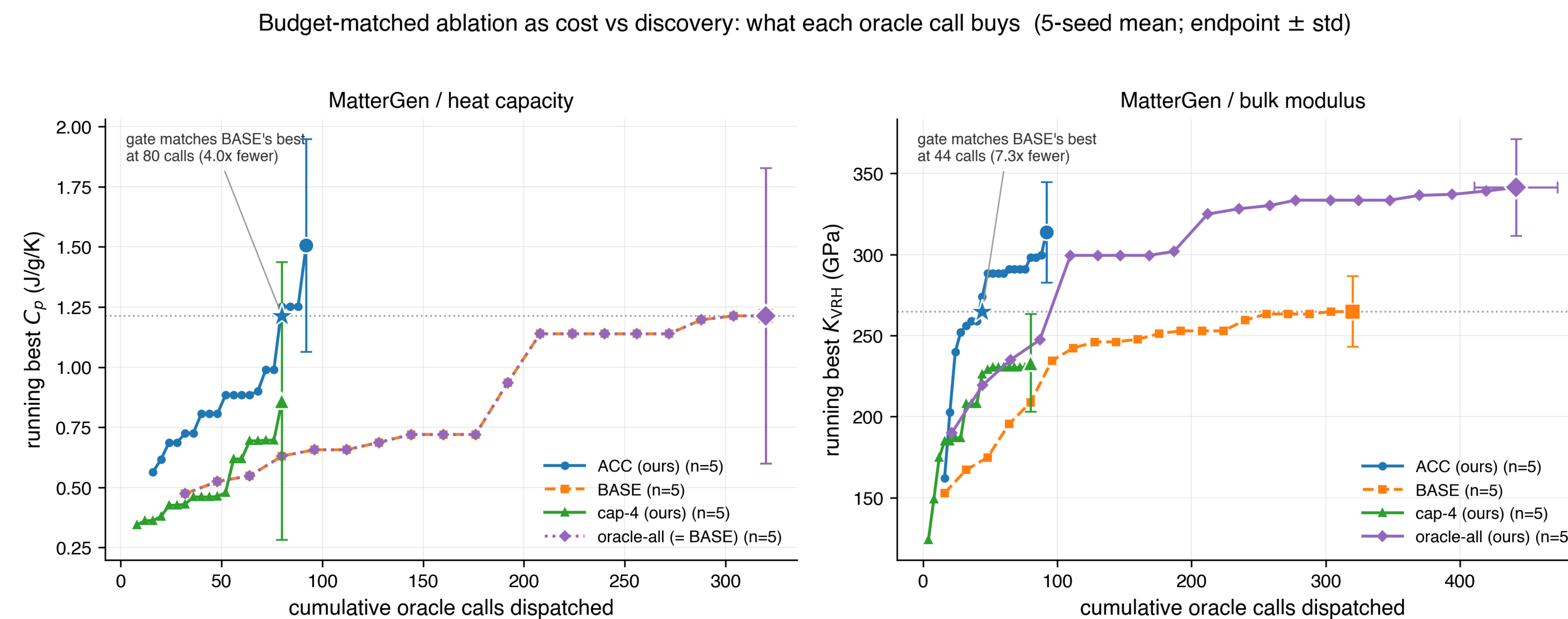
Across mechanical, electronic and vibrational datasets, pretrained ORB embeddings give the highest held-out accuracy; with a Gaussian process on 50 principal components they define the gate.

Discovery Against Oracle Cost



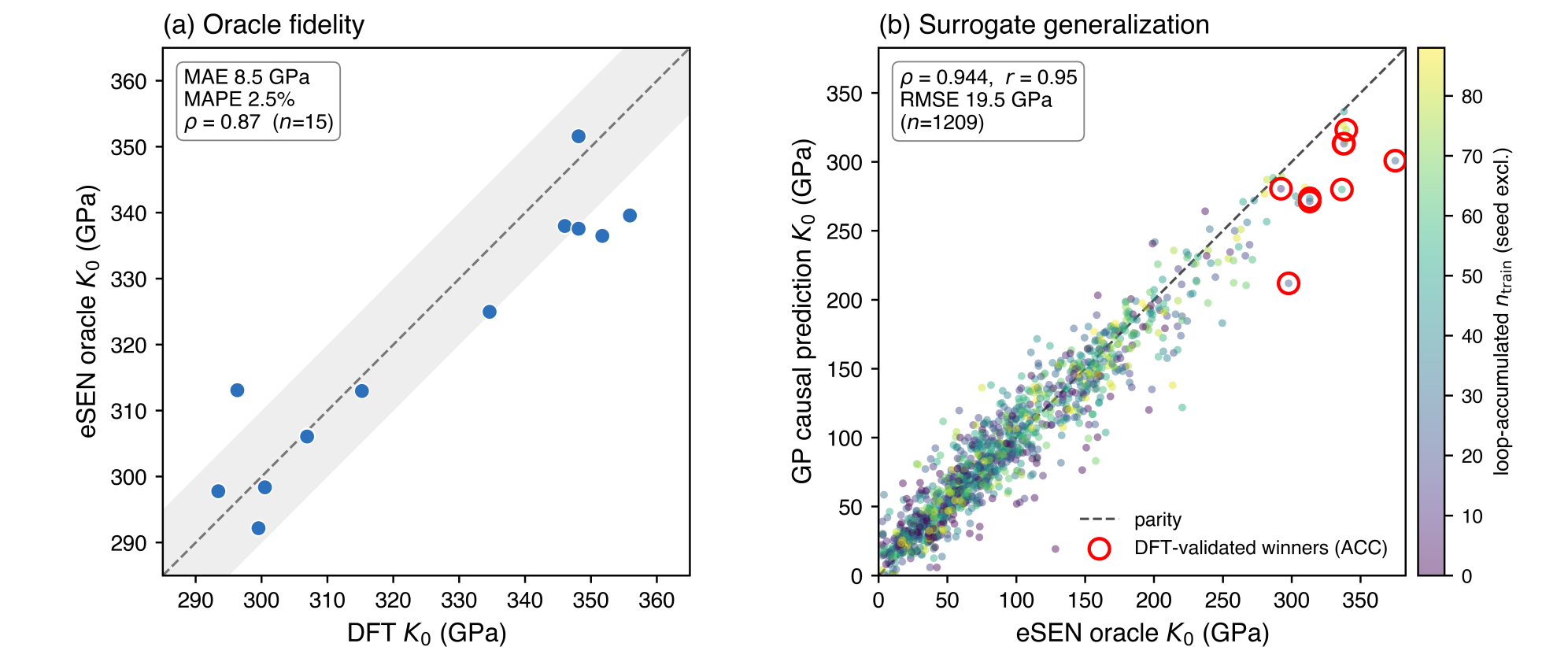
Running best property against cumulative oracle calls for MatterGen, CrystalFlow and ADiT, heat capacity (top) and bulk modulus (bottom), five seeds per cell. The gated policy (solid) reaches the ungated end-of-run best (star) with 2.1–7.3 times fewer oracle calls in all six cells, and caps the budget at 92 calls per 20-cycle run.

Selection, Not Budget, Drives Discovery



MatterGen with a fixed generation front end. Four arbitrary survivors per cycle (cap-4) reach 233 GPa, against 314 GPa when the Gaussian process chooses the four. Sending every survivor to the oracle reaches 341 GPa at 4.8 times the cost: the gate comes within 9% of exhaustive search at about one fifth of the calls.

DFT Validation of Bulk Modulus



(a) On 15 top structures recomputed with DFT, the learned oracle agrees to 2.5% (MAE 8.5 GPa). (b) Replayed causally, the Gaussian process ranks 1,209 unseen generated structures with Spearman $\rho = 0.94$, while underpredicting the hardest extremes: it works as a ranker.

Novelty

None of the top structures match the warm-start pool under strict structure matching (0 of 87 for heat capacity, 0 of 90 for bulk modulus). The best bulk-modulus hit is an MoN polymorph in $P\bar{6}m2$ at 375 GPa (354 GPa in DFT).

Conclusions

- A Gaussian-process gate matches or exceeds ungated fine-tuning while holding oracle calls to a fixed budget.
- The gain comes from ranking-based selection, not from spending fewer calls.
- ORB embeddings with a Gaussian process are the most reliable surrogate across property classes.

References

Chen *et al.*, MatInvent, arXiv:2511.03112 (2025). Zeni *et al.*, MatterGen, *Nature* (2025). Luo *et al.*, CrystalFlow, *Nat. Commun.* (2025). Joshi *et al.*, ADiT, arXiv:2503.03965 (2025). Neumann *et al.*, Orb, arXiv:2410.22570 (2024). Fu *et al.*, eSEN, arXiv:2502.12147 (2025).

Acknowledgements

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Code



Checkpoints